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LapLam: Upcycling panel-shaped wood production waste into larger timber components

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ABSTRACT: The construction sector is a significant emitter of *Greenhouse Gases* (GHG), while also consuming nearly 50 percent of all extracted resources, and producing an estimated 33 percent of waste in Europe. Although building with timber can contribute to reducing the environmental impact of buildings, increasing the use of timber in building structures also leads to more wood production waste. This research investigates the potential of upcycling plate shaped wood waste such as off-cuts or cut-outs into multi-layered building components, such as wall panels, light- and sound diffusers. More specifically, it studies the potential of applying principles inherent to lap-joints in order to achieve coherence in such upcycled timber products. In this paper, we demonstrate how digital tools and workflows, such as *Computer Vision* and various algorithms, can contribute to achieving this. As part of a prototypical workflow, we show how *Computer Vision* can be used to digitize the geometry of a given stock of wood panels and save it to a database. Based on this stock, we identify the challenges that need to be addressed in a custom-made packing algorithm, which will be further developed as part of future studies.

1 INTRODUCTION

Considering its current role as a major emitter of CO_2 and producer of waste, and the *EU*'s goal of becoming a climate neutral economy and society by 2050 (Council of the European Union, 2024), the building construction sector is under a high innovation and transformation pressure. By implementing principles of *Circular Economy* (CE) (Leising et al., 2018), for example by activating wood production waste as a material resource, the amount of waste and CO_2 produced as part of construction could be reduced. Activating production waste as a resource is equally relevant when timber is used as construction material, as wood is a renewable but not inexhaustible resource. This is all the more important when taking into account the increasing use of timber and *Engineered Wood Products* (EWPs) in building construction, which is resource intensive while increasing the amount of wood waste. In Denmark, for example, 30.000 tons of wood waste were incinerated in 2022 overall (Danmarks Statistik, 2022).

In this context, this research investigates the potential of upcycling plate shaped wood production waste such as off-cuts or cut-outs into larger building components, such as wall elements, light and/or sound diffusers. More specifically, it studies the potential of applying principles inherent to lap-joints, which typically are used in combination with linear timber elements in order to achieve longer components (Zwerger, 2015). *LapLam* components consist of multiple layers of panel shaped offcuts, where the positions of elements of adjacent layers are shifted in the horizontal plane, to create an overlap (Hudert, 2024). The overlapping areas will be used to establish coherence between the individual elements, either via adhesives, mechanical fasteners, or both. This will be part of future studies. Among other things, the work is motivated by the assumption that while using the same stock of waste wood elements, varying the size of the gaps could enable

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the assembly of differently sized components. Furthermore, the variation within the used waste could result in the formation of compelling patterns.

Hence, *LapLam* panels also address tectonic aspects, as the appearance of the upcycled panels will not only be affected by the visual and structural qualities of the initial stock, but also by the way in which they are connected, for example with mechanical fasteners and/or adhesives. Future studies will test scenarios for which a panel is designed, for example as a structural or non-structural wall, the respective structural requirements will determine the amount of overlapping between the elements of different layers. The expected load-case could also have an impact on the pattern in which the elements are organized. Obviously, all of the above is only relevant if the upcycled panel remains visible in its built-in stage. By investigating the potential of upcycling panel-shaped wood production waste, by means of restructuring smaller elements to form larger multi-layered panels, this research directly addresses the theme of *ICSA 2025*.

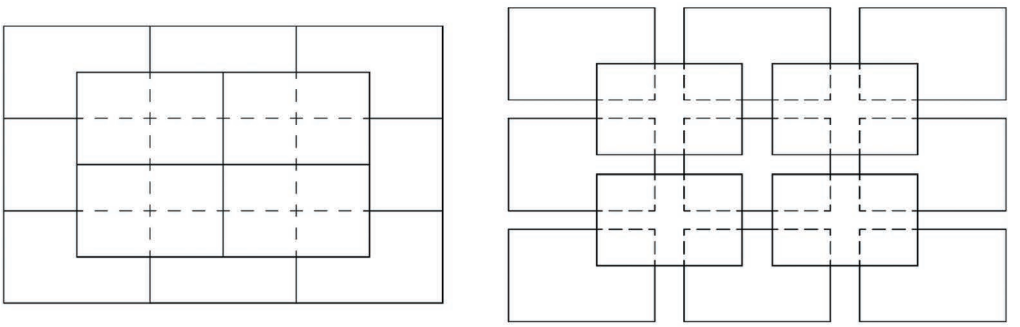


Figure 1. The *LapLam* principle, inspired by lap-joints, could enable the adjustment of the overall size of upcycled components to given dimensions on the construction site.

1.1 *State-of-the-art*

Large parts of current research on circular construction or upcycling of *Construction and Demolition Waste* (CDW) emphasize on how to work with linear steel or lumber members (Huang et al., 2021; Brütting et al., 2021; Warmuth et al., 2020), converting lumber off-cuts into linear beams (Pedersen et al., 2024) or reusing concrete rubble in new wall elements (Grangeot et al., 2024).

Digital tools and workflows play an important role in the upcycling of waste or reclaimed material. In order to process waste or other unconventional materials digitally, they first need to be registered. This challenge could be addressed with scanning equipment, as exemplified by Aagaard & Larsen (2021), which used *3D* scanning to capture the geometry of crooked timber elements. However, this required a significant amount of geometric rationalization. Other industry solutions, such as the 'Woodeye Crosscut' utilize visual information from high-speed cameras to compute cutting sequences in order to avoid knots or other material imperfections in nominal lumber (MicroTec, 2024). In this research, we strive for a lean method, based on *Computer Vision* (CV) and inexpensive web cameras, whereby the two-dimensional data of upcycled wood panels can be registered and saved in a database.

An increasing number of research articles studies the topic of upcycling wood waste, underlining its relevance. Many of these studies (Browne et al., 2022; Ruan et al., 2022; Bergsagel and Heisel, 2023) investigate the potential of reclaimed or salvaged timber, which refers to waste wood obtained from existing and to be demolished buildings. In this case, the elements typically are linear, e.g. planks, boards, or posts. Studies on production waste and plate shaped material mainly look into *CLT* (Poteschkin et al., 2019; Mangliár and Hudert, 2022; Vessby et al., 2023; Dupas and Hudert, 2024). Existing studies on smaller off-cuts (Reisach et al., 2023) investigate the potential of upcycling them into arch-shaped structures. Little if any research is studying the upcycling of plate shaped production waste into larger panels.

Some challenges in this research are similar to those addressed in packing or nesting algorithms. An overview of two-dimensional packing problems is provided in Guo et al. (2022). Existing packing algorithms typically aim for the highest possible packing density and focus only on

the geometric boundaries of the elements to be nested, respectively packed. For upcycled multi-layered timber elements, we also need to consider the grain direction as well as the position and orientation of the elements relative to each other. In contrast to conventional packing algorithms, the relative position of the elements is given. Also, the positioning of elements is not only relative to other elements on the same layer, but has to consider elements on adjacent layers as well.

When addressing packing problems, one can differentiate between off-line and on-line packing, definitions of which are for example provided by Ali et al. (2022). With off-line packing, "full knowledge about items is available beforehand." When using on-line packing, on the other hand, "items arrive one by one and should be packed immediately without having full prior knowledge about them." In this paper, we are applying an off-line approach. On-line packing will be investigated in future studies.

2 DIGITAL WORKFLOWS FOR THE PRODUCTION OF LAPLAM COMPONENTS

In the following, we show how digital tools could facilitate the production of components made from wood production waste, working towards a *LapLam* component that could serve as a non-loadbearing interior wall. We present a technical framework (*Figure 2*) and a computational method (*Figure 4*) to digitize uniquely shaped off-cuts, with the aim of using them in a multi layered panel structure, where the positions of elements of adjacent layers are shifted in the horizontal plane, to create an overlap. This is based on principles found in lap-joints (Zwerger, 2015), typically used to join linear timber elements, where overlapping areas establish coherence between the individual elements, either via adhesives, mechanical fasteners, or both.

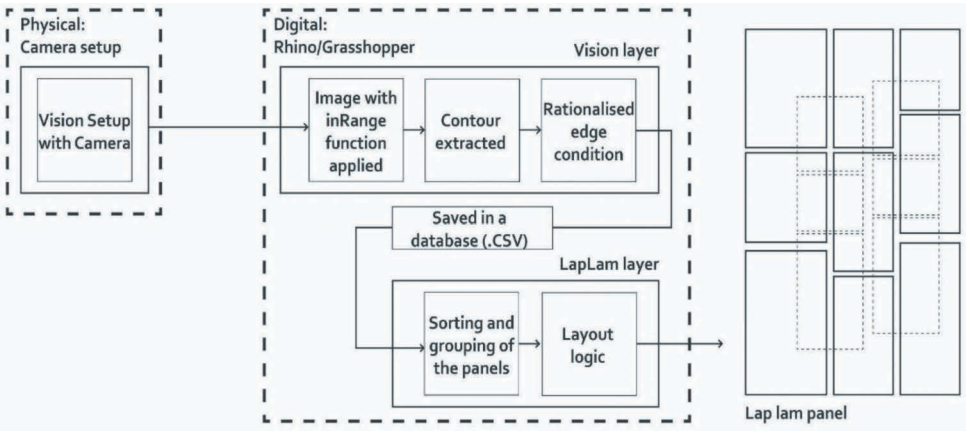


Figure 2. An illustration describing the steps of the developed method, and where the techniques reside: from *CV* system to database, to input data for the design system, which produces a design output and a general process for making and handling resources.

The technical framework consists of two computational layers that have been prototyped in the *CAD* environment *Rhinoceros3D* (Rhino) and its parametric modeling extension *Grasshopper3D* (Grasshopper). The initial layer scans the material stock and stores it in a material database, the second layer distributes the material according to the described *LapLam* logic. The technical framework is demonstrated through a digital case-study, a non-loadbearing wall section. The method is explained in detail throughout the following sections: *Section 2.1* details the resource, section *Section 2.2* details the scanning and storage method, *Section 2.3* addresses the data rationalization, before *Section 2.4* details one way of distributing the material from the data-base into *LapLam* elements.

2.1 *The resource*

To demonstrate the method, the research uses 4 *mm* oak hard-wood, which is either an off-cut or has been discarded due to material imperfections, e.g. cracks and knots, or odd shapes (*Figure 3*). The oak material is normally used for laminate flooring, and has been donated from an industry collaborator, *Dinesen Floors A/S*. The collaborator has supplied a pallet of oak sheets, from which 52 pieces were selected to establish an initial material stock for this research. The small panels have been found to vary in dimensions ranging from 89 to 253 *mm* in width and 166 to 372 *mm* in length. These values are obtained by measuring the distance between the mid-points of opposite edges.

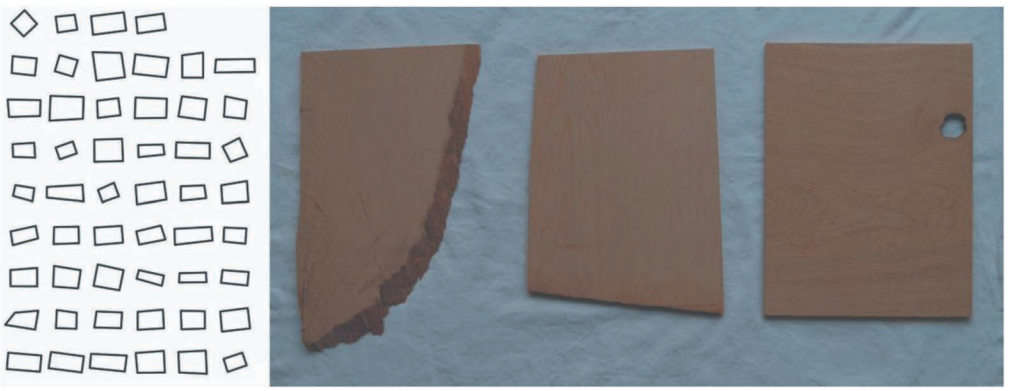


Figure 3. Overview of the chosen stock, consisting of 52 pieces of oak wood offcuts (left), and three exemplary pieces that show variations and imperfections (right).

2.2 *The capturing method*

The capturing method has been prototyped within *Rhinoceros* and *Grasshopper*, where a plugin has been written in C#.net and uses the *EmguCV* library, which is a *C#.net* wrapper of the commonly used *OpenCV*. The plugin learns from Pedersen et al., 2021, to establish a method to calibrate the camera, computing the scale of objects within an image, and how to isolate objects within a camera stream based on material color. As a result, this research follows the described steps, and continues to use the 'inRange' function to convert images to alpha maps (white/black images) (*Figure 4*, C), which allows to extract the outline from isolated objects within the resulting alpha maps, using a contour function of *EmguCV* (*Figure 4*, D). The extracted contour is complex and jagged, and is rationalized, using the *Grasshopper* native genetic algorithm solver, into a simplified polygonal representation (*Figure 4*, E-F), before being saved into a *.CSV* document that contains the bounding box's width/height and coordinate positions of the rationalized polygon.

The physical setup is important for the method, of the capturing method consists of a *Logitech c270* webcam, a table and a white sheet (*Figure 4*, left). The camera is positioned on the table and is facing the floor, where the white sheet creates a contrasting background to the oak panels. In this study, capturing the contour of the elements is sufficient, as the thickness of the elements is known to be identical. Although identifying the grain direction is crucial in the production of material aware and potentially structural upcycled components, it is not implemented in this study. It will be addressed as part of future research.

2.3 *Data rationalization*

The panels were captured in random sequence and orientation, which resulted in unstructured data. Therefore, the research used a *k*-Means algorithm (Hartigan & Wong, 1979) to structure the panels into specified groups, which makes it simpler to use them as part of the *LapLam* logic. The *k*-Means algorithm uses the width or height of the bounding box, to place the panels into groups of similar size domains, see *Figure 5*. Additionally, the panels were digitized with a random orientation, and therefore reoriented to match the *x* axis.

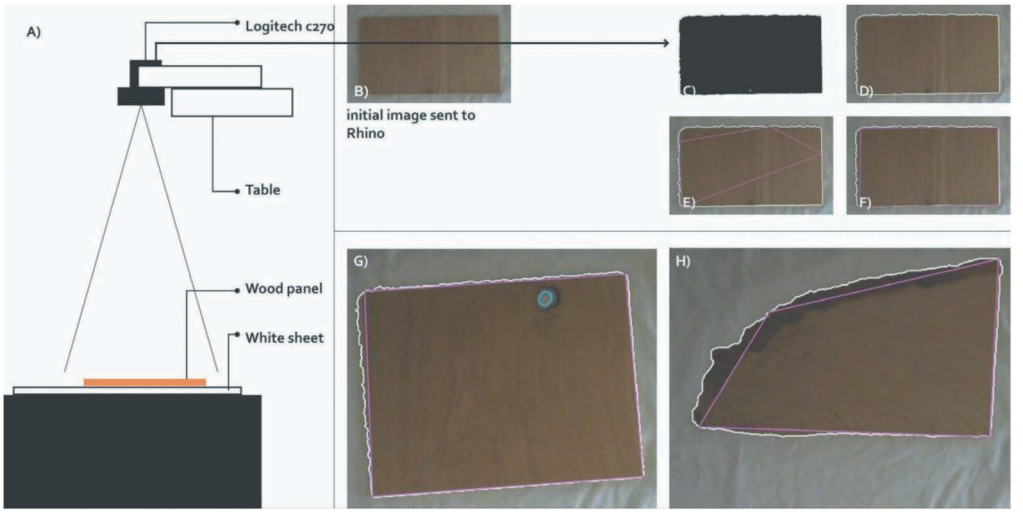


Figure 4. A) Describes the physical setup, and subsequent images. B-F Display the steps of generating the rationalized outline (pink). Lastly, images G-H show two additional panels and their rationalized outline.

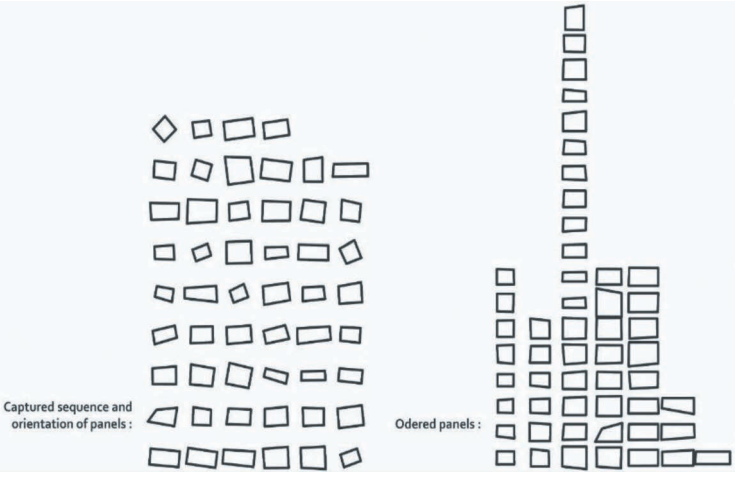


Figure 5. This diagram shows the effects of reordering and grouping the panels based on its associated data. Within this diagram panels are sorted using the width and is organized into seven groups.

2.4 *Packing strategies for LapLam components*

Different strategies can be considered for the production of *LapLam* panels, which involves positioning the elements of a chosen stock within the boundaries of the component to be achieved, and positioning elements of one and adjacent layers relative to each other, according to certain constraints and goals. Possible strategies for achieving this include packing algorithms, or spring systems. Another option for packing panels could be *Graph Recurrent Neural Networks* (GraphRNN), as shown in Silkeborg and Christensen, 2024 (BSc thesis, Aarhus University, 2024). As part of this paper, we propose the implementation of a spring system, for example using the *Kangaroo* solver, which is part of the visual programming environment *Grasshopper*. The constraints are the boundary of the target geometry and the overlap of panels of adjacent layers. The goal is to equalize the distance between the panels. A possible workflow is described in the following, and illustrated in *Figure 6*. The starting point is a defined target

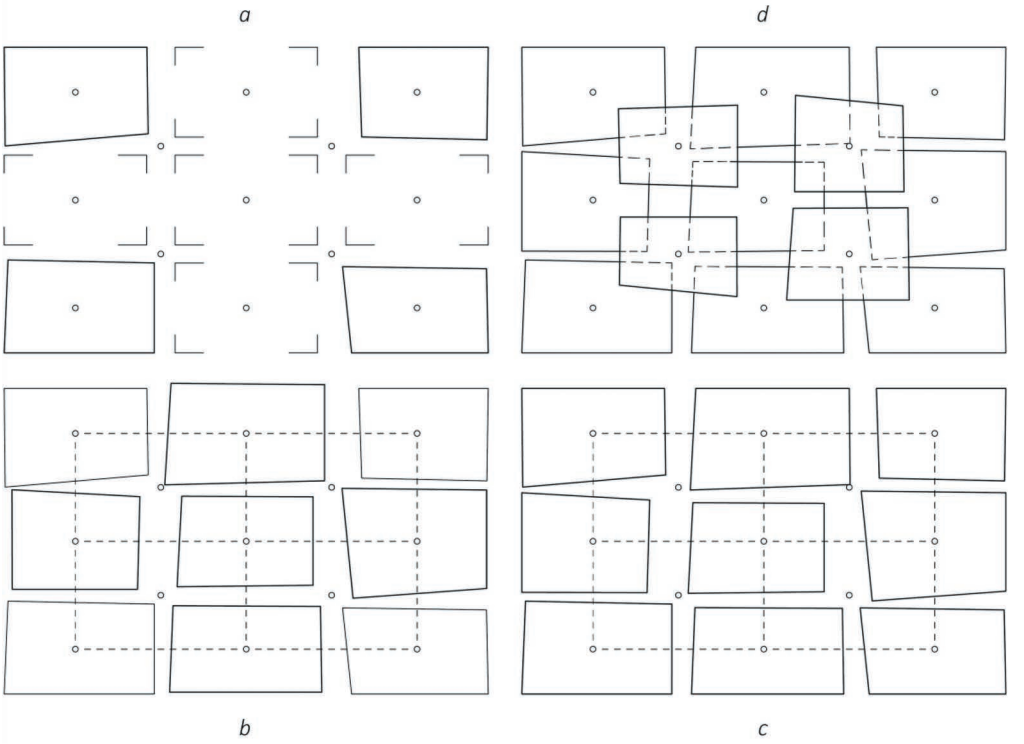


Figure 6. Four stock elements are placed and anchored in the corners of the target geometry (a). Five elements are placed in-between them, based on a grid derived from an idealized layout (b). A spring system equalizes the distances between the elements (b, c). The second layer consists of four elements that are placed in-between those of the first layer (d).

geometry, respectively the desired dimensions of the resulting *LapLam* panel. The dimensions of the target geometry are subdivided by the stock elements' average width and length to generate an initial idealized distribution pattern.

First, four elements are positioned and anchored in the corners of the virtual boundary of the target geometry, such that one or two neighboring edges of each elements are aligned with the boundary lines (*Figure 6a*). Elements are picked from adjacent size domains, to avoid too large differences in size. The five remaining elements are placed on points taken from an idealized configuration, using the average element length and width as starting point. Following that, a spring system is implemented, equalizing the distances between the panels (*Figures 6b and c*). The second layer consists of four panels that are placed such that each of them creates overlaps with four panels of the first layer (*Figure 6d*). The third layer follows the logic of the first one.

3 RESULTS, DISCUSSION, AND CONCLUSION

Based on the *LapLam* principle, this research set out to investigate suitable digital workflows for the production of multilayered upcycled timber components. Using *Computer Vision*, size domains, and a computational packing strategy based on a spring system, we developed a prototypical workflow to achieve this. In the process, we have shown that conventional packing algorithms are relevant, but not fully equipped for organizing waste wood in a material aware manner. The proposed workflow is specifically designed for non-loadbearing *LapLam* panels with gaps, and limited to wood production waste. Additional challenges emerge when looking into reclaimed wood. Further work is required to scale up the here presented results and to achieve full-scale building products. In *Figure 7*, we provide schematic renderings of a possible *LapLam* panel configurations, with either nine or four elements on the outer layer.

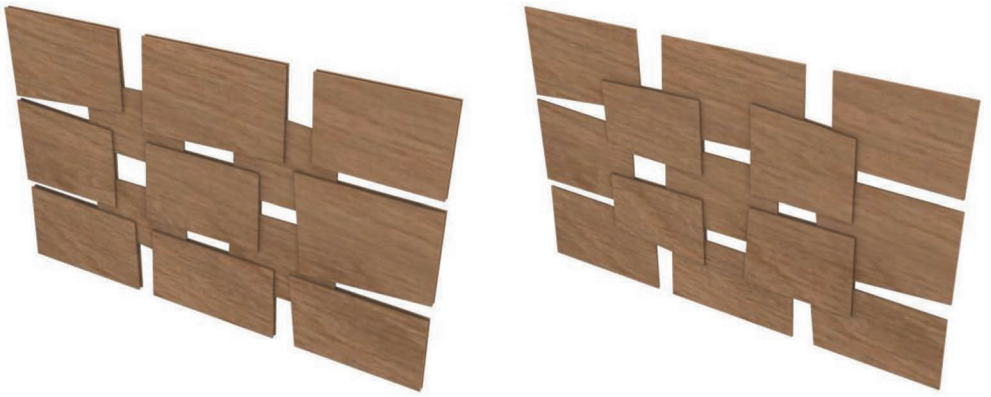


Figure 7. Schematic renderings of possible *LapLam* panel configurations.

4 OUTLOOK

Based on the here established workflows, we plan the prototypical production of various configurations of *LapLam* panels with the help of a *UR5e* robotic arm. In addition to that, we will study alternative methods to the spring-system approach, taking advantage of the proposed domain size grouping, as well as bottom-up approaches. In order to achieve full-scale prototypes, an industrial robot, or other large-scale infrastructure, is required. Another planned step is the identification of the elements' grain direction and considering it in the composition of *LapLam* panels. This would be particularly relevant in the case of loadbearing applications. In this context, it could also be relevant to study different types of connectors, such as wooden dowels or screws, potentially combined with adhesives. We intend to study the structural performance of panels with different types of connectors via *Finite Element Analysis* (FEA). Furthermore, we see potential for investigating on-line bin packing, using elements directly as they are digitized. This could reduce the logistical challenges of circular construction, as the need for storing stock elements would be significantly reduced. Lastly, whereas this paper is limited to production waste, it would be logical to expand the scope, and consider reclaimed wood as well. When aiming at load-bearing applications, further steps are required, as described for example in Tamke et al. (2024). We also consider the integration of materials such as mycelium or bio-based insulation into *LapLam* components.

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